

COMP 141 – Introduction to Computing Tools and Techniques

Course Syllabus – Spring 2026

COMP 141 uses a common syllabus for all sections this semester. Please check LOCUS (and Sakai) to ensure you know your section and instructor's name.

Course Description

This course introduces the Unix shell environment and command line interface tools. Students who complete it will be fluent in the Unix shell as an instrument for operating computers optimally, helping them succeed in future academic, research, and professional computing disciplines and needs.

Motivation

Despite the current era of rich graphical user interfaces (desktop, web, and mobile), the "command line" experience remains important, especially for software developers and computer-aided scientific researchers. Many development scenarios still require command-line use and fluency in Unix tools, including in modern embedded, cloud, cybersecurity, and supercomputing environments. With mobile computing and the Internet of Things, Unix (notably via Linux) has become a ubiquitous platform thanks to its lightweight and open-source nature. While this course does not aim to be a comprehensive programming class, students will master essential programming skills using shell scripting. They will also learn about computer-aided problem-solving using Unix commands supported by shell scripts.

Prerequisites

There are no prerequisites.

Course Delivery

The course material is fully online, with asynchronous lecture videos and readings, and weekly synchronous in-person (section 002) or online (section 003) meetings for students to ask questions and get content clarification from the instructor. We will use Sakai (sakai.luc.edu) as the online platform to support, centralize, and store the course announcements, materials, activities, assignments, quizzes, progress reports, and grading communication.

Sections

Section	Sync Class Time	Instructor	Classroom or Link	E-mail @luc.edu
002	Tu 4:15 PM – 5:30 PM	George Thiruvathukal	Dumbach Hall 234	gthiruvathukal
003	We 4:15 PM – 5:30 PM	William O'Connell	Online	woconnell

Teaching Assistants

All the sections will share the following teaching assistant (TA). Please feel free to contact them with questions about the course content and help with the assignments.

- Ms. Ale Vellucci Solari avelluccisolari at luc.edu

Office Hours

In addition to the weekly online synchronous meetings, instructors will offer office hours in the following terms.

Section	Instructor	Modality	Appointment Link
002	George K. Thiruvathukal	Online or in-person by appointment	calendly.com/gkthiruvathukal
003	William O'Connell	Online by appointment	TBD

Grading

Every activity matters. There will be one practical **assignment** around every other week. Short assignments will be worth 5 points each, while long assignments will be worth 10 or 15 points each. An online multiple-choice **quiz** will be worth 10 points around every four weeks. The final exam will be an online multiple-choice quiz open during finals week.

The final score will be the total points earned divided by the total points possible. This score will be converted to a concept grade according to the following scale.

Grade	Score	Grade	Score
A	>= 93%	C+	>= 77%
A-	>= 90%	C	>= 73%

B+	>= 87%	C-	>= 70%
B	>= 83%	D+	>= 67%
B-	>= 80%	D	>= 63%
		F	< 63%

Late Work and Resubmission Policy

Late work submission is accepted but subject to the following penalties.

Days Late	0	1-2	3-6	7-13	>= 14
Score Deduction	0%	5%	10%	20%	100%

Late penalties may be waived upon presentation of appropriate documentation justifying the delay.

Resubmission is allowed any number of times before the work deadline.

Makeup Policy

Activities such as quizzes missed without a pre-excused absence will be graded as zero. The only exception to this rule is a documented medical or health condition that prevents the student from providing prior notification.

The instructors reserve the right to give a makeup quiz completely different from the regular one taken by the other students. It may include arranging an alternative activity that requires a quiz makeup outside regular class time, with a short turnaround time.

Course Topics

The course content is heavily based on William Shotts' book "The Linux Command Line", freely available online at linuxcommand.org/tlcl.php. There are also additions made by the instructors. The covered topics are the following:

1. Preparing your computer – or a remote one – for operation: MacOS Terminal, Windows Subsystem for Linux, or Ubuntu Linux running natively.
2. Command-line operation: calling programs, and passing options and arguments.
3. Time-saving practices and notations: keyboard shortcuts and regular expressions.
4. Filesystem: files, directories, user and group permissions (chmod, chown), file and directory manipulation (cp, mv, rm, cat, mkdir, cd, ln, etc.).

5. Processes: running and monitoring programs, job, foreground, and background control, I/O redirection, stream commands (e.g., cat, grep, head, tail, tee).
6. Shell environment: reading and setting variables up, significant system variables (e.g., PATH, USER, HOME), and prompt customization.
7. Text editing: fast and advanced operation with Vi.
8. Program installation: package management, software installation, removal, and update.
9. Networking: remote access, file transferring, and examination of network configuration.
10. Archiving and file compression: usage of tar, zip, gzip, etc.
11. Text processing: usage of commands to quickly read and manipulate file content, such as cat, sort, uniq, cut, paste, join, diff, sed, etc.
12. Program compilation and interpretation: compiling programs in C and Java, and using interpreted languages such as Python.
13. Shell scripting: programming basics to demonstrate the potential of automating tasks in the shell.
14. Version control: Git profile creation, basic usage, and source code version control.

Tentative Schedule¹

Week	Topics	Assignment	Quiz
Week 1 (Jan 12)	Environment Setup, What is the Shell?	A1	
Week 2 (Jan 19)	Navigation, Exploring the System		
Week 3 (Jan 26)	Manipulating Files and Directories, Working with Commands	A2	
Week 4 (Feb 2)	Redirection, Seeing the World as the Shell Sees it		Q1
Week 5 (Feb 9)	Advanced Keyboard Tricks, Permissions	A3	
Week 6 (Feb 16)	Processes, The Environment		
Week 7 (Feb 23)	Customizing the Prompt, Package Management	A4	
Week 8 (Mar 2)	<i>Spring Break</i>		
Week 9 (Mar 9)	A Gentle Introduction to Vi, Networking		Q2

¹ Dates indicate the beginning of each week. Specific class dates depend on your section.

Week 10 (Mar 16)	Searching for Files, Archiving and Backup	A5	
Week 11 (Mar 23)	Regular Expressions, Text Processing		
Week 12 (Mar 30)	Compiling Programs, Writing Your First Script	A6	
Week 13 (Apr 6)	Version Control and Git, Terminal Multiplexing		Q3
Week 14 (Apr 13)	Keyboard Input with "read", Positional Parameters	A7	
Week 15 (Apr 20)	Branching with IF, Looping with FOR, WHILE, and on Files		
Week 16 (Apr 27)	<i>Finals Week</i>		Q4

Learning Outcomes

- Acquiring skills required for success in high-end computing jobs and graduate studies.
- Mastering the Unix philosophy, which has lasting benefits when it comes to software development. By writing tools that are focused and do one job well, students will write software with fewer bugs that is more likely to stand the test of time. The Unix philosophy draws on pioneering ideas from other great thinkers, such as Mies van der Rohe, who advocated “less is more” as a fundamental design principle.
- Understanding that some traditions of computing are worth the investment of time and should be carried forward. Understanding Unix is helpful in understanding the remainder of the Computing disciplines. For instance, ideas such as regular expressions (first found in Unix) are wired into every modern programming language.
- Learning a text editor (Vi/Vim) remains important when working on remote servers, especially in supercomputing and embedded systems. Vi/Vim is not the only choice, but it is closely aligned with the university’s social justice system and has one of the most vibrant communities. The creator of Vim works to improve the conditions of refugees and children in Uganda!
- Being more confident in other introductory courses. Knowing how to edit and compile programs, write scripts, and work with Unix-like systems allows us to focus more on the ideas of computer science and programming fundamentals rather than fighting with the computer. Because every student will learn to operate a computer in a more advanced and conscious way, the techniques learned in this course can and will be carried forward to all courses, even when students choose to use Windows (which added Linux support as a subsystem). The fact that Linux, MacOS, and Windows all offer Unix-like semantics means the command line is important to them and their business interests.
- Acquiring skills needed to succeed in systems, DevOps, and cybersecurity domains, where this course’s tools are crucial to problem-solving.

- Prepare students for the “hard” CS programming classes, e.g., programming languages, operating systems, distributed systems, high-performance computing, etc., which assume fluency with the command line.

Academic Honesty

Students are expected to read the statement on academic integrity, available at catalog.luc.edu/academic-standards-regulations/undergraduate/undergraduate.pdf#page=3.56.

This policy applies to this course. The minimum penalty for academic dishonesty is an F on that assignment. Multiple instances or a single severe instance on a major exam or assignment may result in a grade of F for the course. All cases of academic dishonesty will be reported to the department office and the relevant college office, which will place them in your school record.

Academic dishonesty includes, but is not limited to, working together on assignments that are not group assignments, copying or sharing assignments or exam information with other students except in group assignments, submitting as your own information from current or former students of this course, copying information from anywhere on the web and submitting it as your own work, and submitting anything as your own work which you have not personally created for this course. If you wish to use materials that are not your own, please check with the instructor beforehand and cite your source clearly. When in doubt, ask first! Be aware that the instructors may have changed the values for many textbook problems used for homework problems. For those problems that require open-ended answers, please be very careful to state the answers in your own words, not in the words of the instructor's manual, nor in the words of students who have previously taken this course.

Privacy Statement

Ensuring privacy for faculty and students engaged in online and face-to-face instructional activities helps promote open and robust conversations. It mitigates concerns that comments made in class will be shared beyond the classroom. As such, recordings of instructional activities occurring in online or face-to-face classes may be used solely for internal class purposes by the faculty member and students registered for the course, and only during the period in which the course is offered. Students will be informed of such recordings through a statement in the course syllabus. Instructors who wish to make subsequent use of recordings that include student activity may do so only with informed written consent of the students involved or if all student activity is removed from the recording. Recordings including student activity initiated by the instructor may be retained by the instructor only for individual use.

Policy for Recording Live Sessions

In this class, software will be used to record live class discussions. As a student in this class, your participation in live class discussions will be recorded. These recordings will be available only to students enrolled in the class, to assist those who cannot attend the live session or to serve as a resource for those who would like to review the content presented. All recordings will

no longer be available to students in the class after the Sakai course is unpublished (i.e., shortly after the course ends, per the Sakai administrative schedule). Students who prefer to participate via audio only will be allowed to disable their video camera so only audio will be captured. Please discuss this option with your instructor.

Religious Holidays

If you need to miss class for a religious holiday, please alert the instructor in advance so that it will not be counted as an absence. As required, due dates for quizzes and tests, homework, and assignments can be adjusted if they conflict.

Students with Disabilities

Loyola University Chicago provides reasonable academic and student life accommodations for students with disabilities. Students interested in receiving academic accommodations related to a disability or other health condition are invited to register with the Student Accessibility Center (SAC) at www.luc.edu/sac/applywithsac. Students may register with the SAC at any point in their academic career. However, the SAC encourages students to register as early as possible to reduce delays in the delivery of accommodations. The SAC, student, and professor will engage in an interactive process to determine how each student's accommodations are applied to individual class sections.

Please note that in this class, software may be used to audio-record class lectures to provide equal access to students with disabilities. Students approved for this accommodation use recordings for their personal study only, and recordings may not be shared with others or used in any way against the faculty member, other lecturers, or students whose classroom comments are recorded as part of the class activity. Recordings are deleted at the end of the semester.

For more information about registering with the SAC or questions or concerns regarding the implementation of approved accommodations, please contact the SAC at 773-508-3700 or SAC@luc.edu. Students are also welcome to visit the SAC on the first floor of Sullivan Center, Suite 117, to share questions or concerns with one of our accessibility specialists.

Continuous Improvement

We believe in a personal quality process of continuous improvement. Anything can be improved by applying the "Plan, Do, Check, Act" (PDCA) quality process. To improve the course and the learning of programming and computer science concepts, we welcome your feedback, comments, suggestions, and complaints at any time.

We may ask you to participate in surveys and discussions during the course to support the PDCA process above. These surveys will measure student impressions of the course; when time permits, we will share the results with the class. Your input on these surveys is anonymous and does not affect your grade.

We will also use a LUC feedback system called IDEA toward the end of the course to gather your feedback on both the course itself and the instructor's teaching. We use this feedback to improve how we teach the course material, and the CS Department uses it to refresh and enhance our course offerings.

Responsible Campus Partners

Instructors are Responsible Campus Partners ("RCP") under Loyola's Comprehensive Policy and Procedures for Addressing Discrimination, Sexual Misconduct, and Retaliation (available at www.luc.edu/equity). While the instructors' goal is for you to be able to engage fully and authentically with the course material through class discussions and written work, the instructors also want to be transparent that as an RCP, they must notify the Office for Equity & Compliance ("OEC")/Title IX Coordinator when they have any information about conduct that reasonably may constitute Title IX Sex-Based Discrimination, as defined in www.luc.edu/equity/titleixequitylaws/titleix.

Moreover, instructors also have an obligation under Illinois law to report disclosures of or suspected instances of child abuse or neglect, as defined in www.luc.edu/hr/legal-notices/mandatedreportingofchildabuseandneglect.

The University maintains such reporting requirements to ensure that any student who experiences sexual/gender-based violence receives accurate information about available resources and support. Such reports will not be submitted to law enforcement (no student will ever be forced to file a report with the police). Additionally, the University's resources and support are available to all students, even if a student chooses not to have any other action taken. If you have any questions about this policy, you are encouraged to contact the OEC at equity@luc.edu or 773-508-7766.

If you ever wish to speak with a confidential resource regarding gender-based violence, we encourage you to call "The Line" at 773-494-3810. The Line is staffed by confidential advocates from 8:30 am-5 pm M-F and 24 hours on the weekend when school is in session. Advocates can provide support, talk through your options (medical, legal, LUC reporting, safety planning, etc.), and connect you with resources as needed – without generating a report or record with the OEC. More information about The Line can be found at luc.edu/wellness.